In-vitro Biology

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Protein Expression Workflow

Protein sciences and structural biology

Gene to structure

Expression System

- Mammalian: CHO, HEK 293, stable pool, single clone and transient
- *E. coli*: Soluble, inclusion bodies
- Insect cells: Baculovirus mediated, Sf9, Hi5 and Super Sf9

Optimization of complex targets

- Structure based construct optimization
- Scouting of multiple constructs, tags and expression systems
- Vector and induction conditions
- Media, feed, seed density etc.

Scale: mg to g

- Mammalian and insect cells: Shake flask, spinner and Wave Bioreactor (25 L)
- *E. coli* and *Pichia*: Shake flasks

Analytical QC

- SDS-PAGE (reduced, non-reduced)
- SE-HPLC
- Mass spec (intact, reduced-deglycosylated, N and C-term confirmation, PMF)
- Endotoxin: Target less that 1 EU/mg)

Bio-conjugation

- ADC development (details in NBE section)
- Bio-Conjugation for immunization and assay reagent development
- Various commercial and customized linkers
- Standard conjugation- Biotinylation, FITC, Ruthenylation etc

Protein structure determination by X-ray crystallography

- Synchrotron diffraction data collection in mail-in format
- Manual crystallization set up (sitting drop, hanging drop and micro-batch) will be performed for unbound and complexed targets using commercial screens.
- Soaking for inhibitor complex generation will also be employed.

Gene to Structure

X-ray crystallography workflow

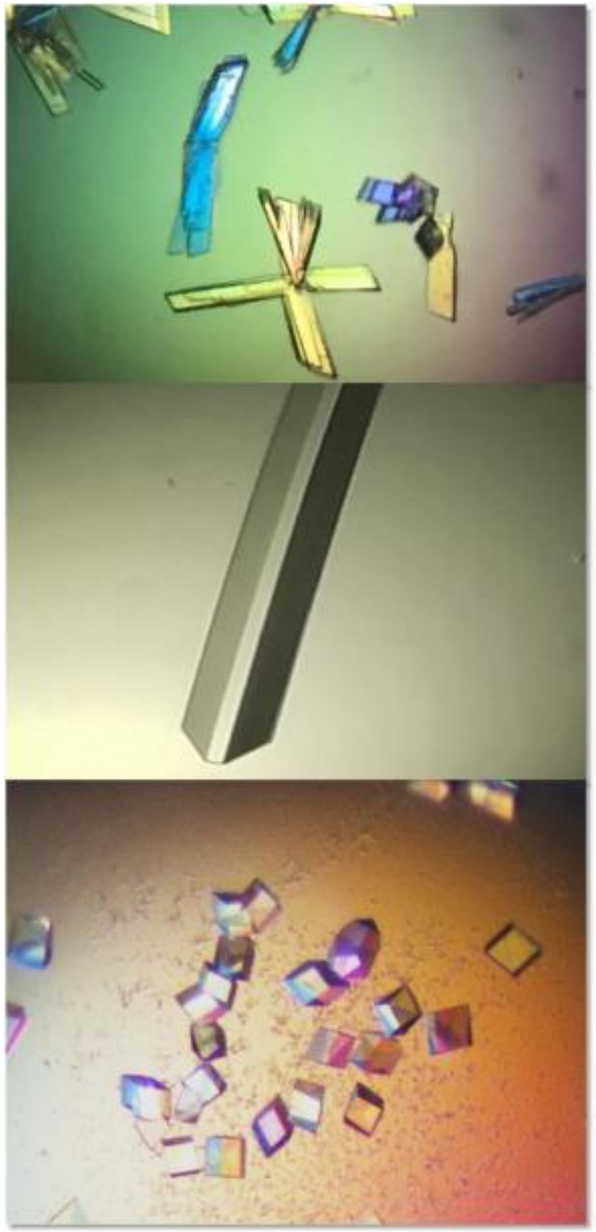
Initial screen
- Commercial screens
- Customized screens
- Sitting drop, hanging drop, microbatch etc.
- Manual set up

Optimization
- Customized expansion
- Seeding (micro and macro)
- Additive screen
- Temperature, pH etc.

Ligand complex
- Protein-ligand complex purification
- Co-crystallization
- Soaking
- Fragment library screen

Crystal ID
- Microscopy
- Staining
- Electrophoresis
- Preliminary X-Ray diffraction

Cryo-protectant screen
- Standard commercial reagents
- In-house cryo-protectants
- Automated screening at synchrotron
- Optimization of cryo-conditions



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Gene to Structure

X-ray crystallography Quality parameters

Protein engineering for better **Crystallizability**

2.5 Å or better resolution
>85% completeness
I/Sig I >1.5 (in highest resolution bin)
20 or better R_{work}
25 or better and R_{free}
Bond angle RMSD 0.02
Bond length RMSD 1.5
>80% Ave correlation co-efficient
Clear electron density of ligand

Synchrotron data collection
Preferred: Australian Synchrotron

Molecular replacement
Experimental phasing
Refinement
Ligand fitting
Structure validation
Structure analysis
Structure file in PDB format

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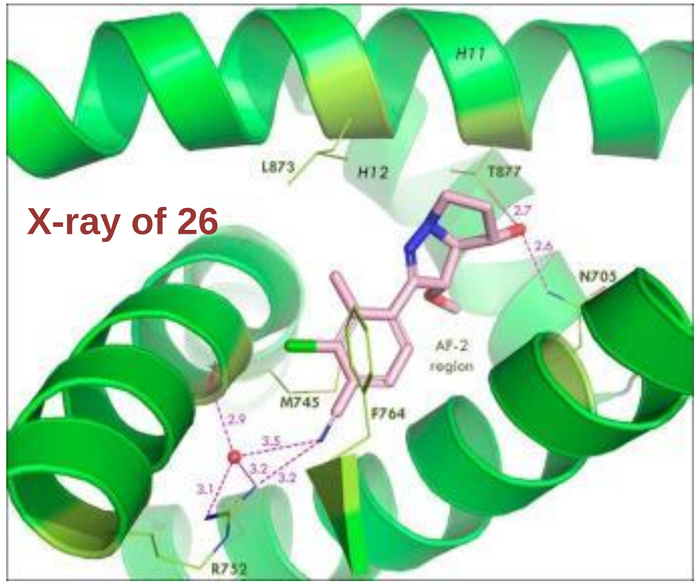
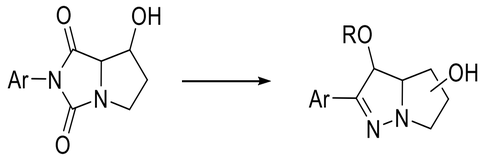
Case study 3: IDD program for Androgen receptor

Structure-aided design of novel SARMs

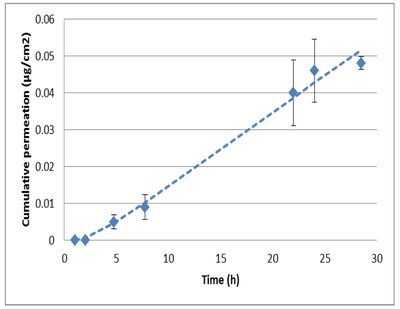


- 3-Alkoxy-pyrrolo[1,2-b]pyrazolines as novel selective androgen receptor modulators (SARMs) with ideal physicochemical properties for transdermal administration
- Structural modelling assisted in understanding tolerability of different substitutions in AF-2 pocket for PK optimization

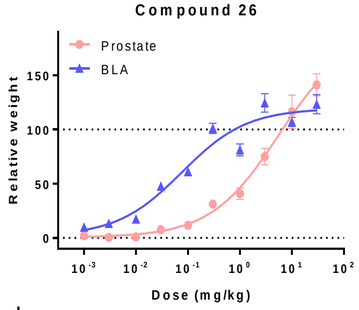
FAST FOLLOWER DESIGN PHILOSOPHY



[5+5] ring hydantoin scaffold → [5+5] ring pyrazoline scaffold



Compound 26 penetrates through human skin from a saturated aqueoussolution buffered at pH 7.4.The observed flux at steady state (Jss) is 2.0±0.2 g/(cm2*h),



Hershberger in vivo assay of 26

Compound	R1	R2	C2C12 EC_{50} [nM]	C2C12 E_{max} [%]
DHT			0.13 ± 0.02	100
21	H	(S)-OH	978 ± 221	55 ± 21
26	Me	(S)-OH	0.5 ± 0.4	97 ± 13
27	Et	(S)-OH	1 ± 0.5	98.3 ± 2.9
28	i-Pr	(S)-OH	4.5 ± 0.1	102 ± 2.8

Ref: J. Med. Chem., 2014, 57 (17), pp 7396–7411

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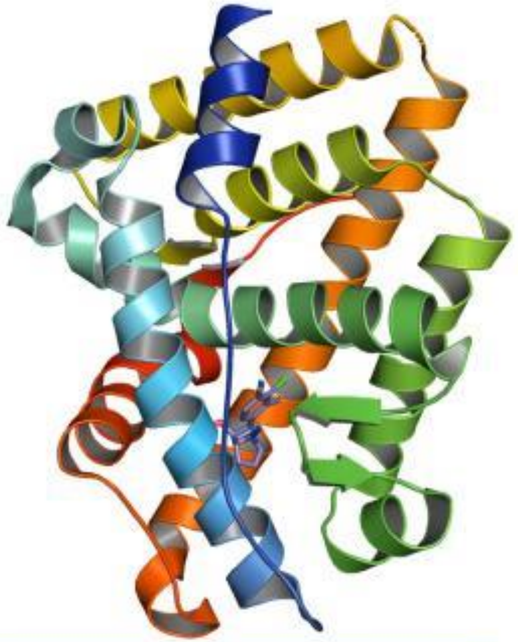
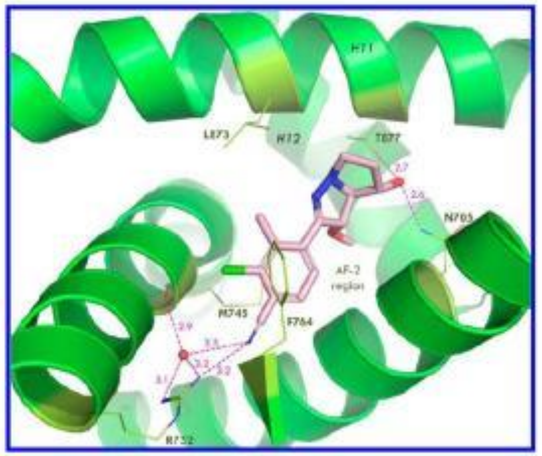
Case study 3: IDD program for Androgen receptor

Structure-aided design of novel SARMs

space group	$P2_12_12_1$
cell parameters (Å)	a = 55.129, b = 65.476, c = 68.964
diffraction resolution (Å)	50.0–2.10
total reflections	42 722
unique reflections	14 439
completeness (%)	96.8 (80.4)[a]
R_{sym}	0.048(0.134)
$I/\sigma I$	19.3 (3.2)
multiplicity	3.5(3.3)
refinement	
resolution (Å)	29.62–2.1
no. of reflections	13 660
completeness (%)	94.73(89.28)
R_{work}/R_{free}	0.245/0.275
mean B-factors	20.403
rms bond (Å)	0.005
rms angles (deg)	0.784





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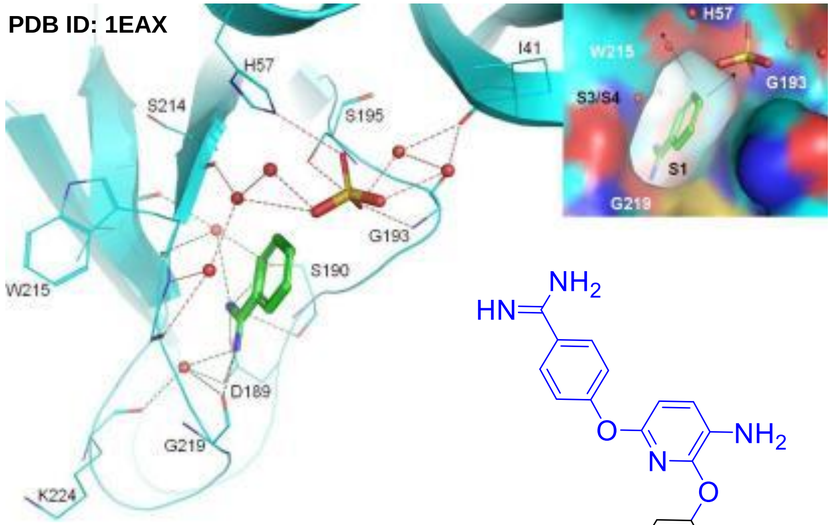
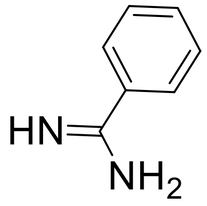
Case study 4

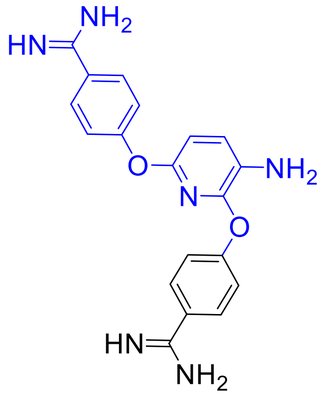
Structure-based design of matriptase inhibitors



- Inspiration: Benzamidine binding as sulfate salt
- Hypothesis: Benzamidine substitution to engage sulfate pocket & solvent exposed S3/S4 region
- WaterMap® to identify distinctive binding hotspot: 100× binding improvement
- In house co-crystal structures validated modeling hypotheses
- Favorable selectivity over undesirable serine proteases, cellular activity, solubility & metabolic stability

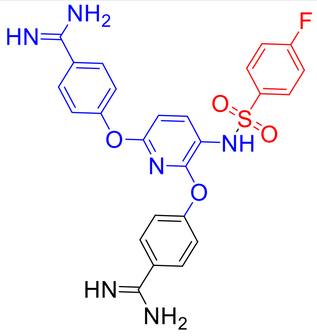


Benzamidine, K_i~ 400 µM



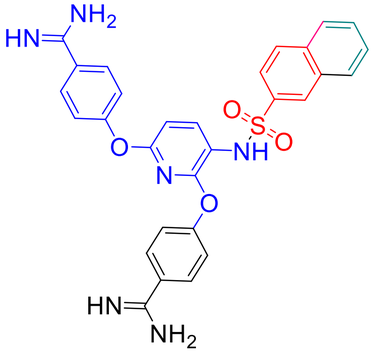
2

K_i = 2.9 µM



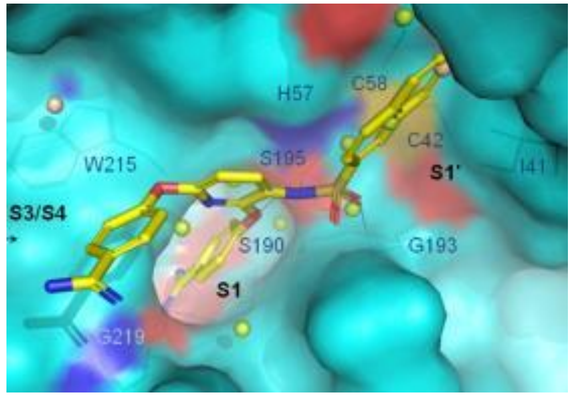
8

K_i= 0.2 µM



10

K_i = 0.04 µM



ACS Medicinal Chemistry Letters **2013** *4* (12), 1152-1157

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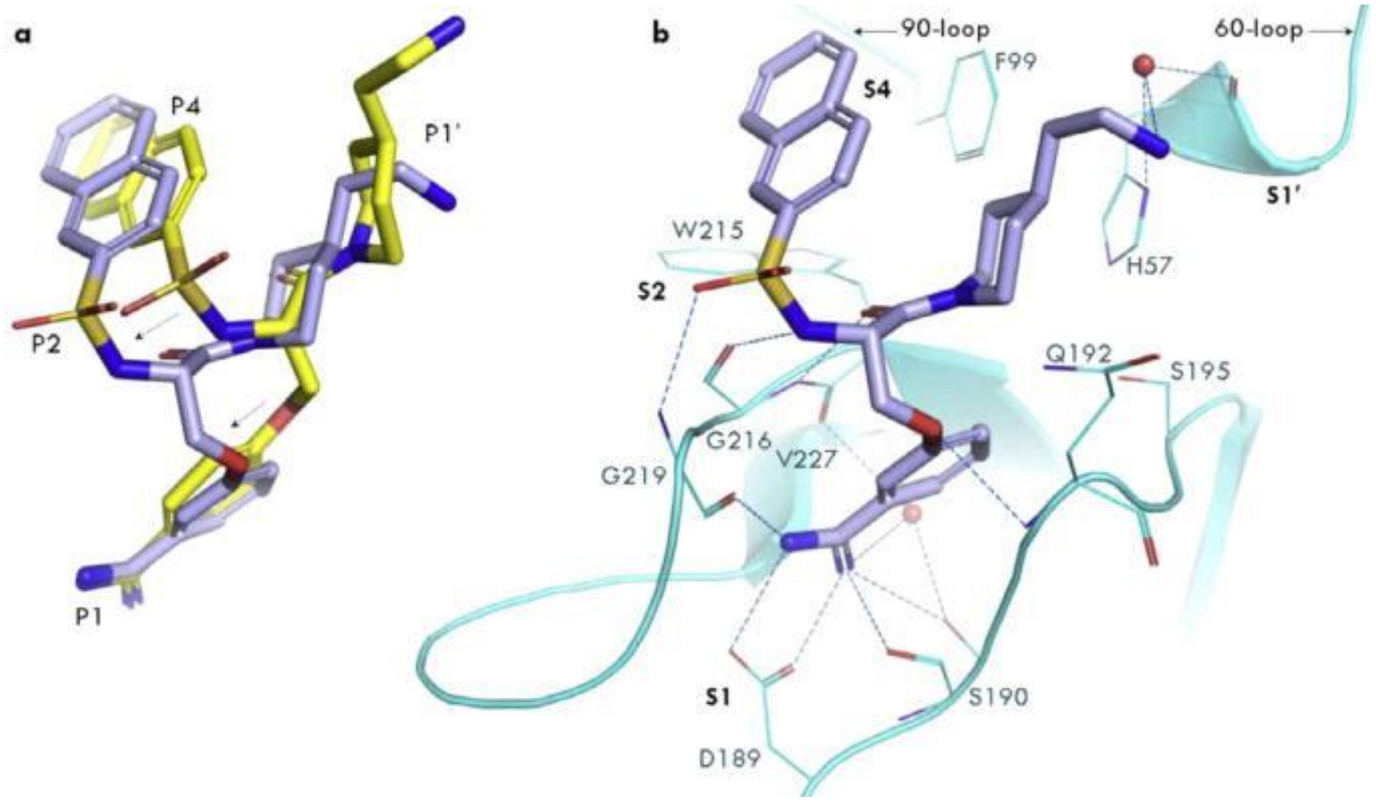
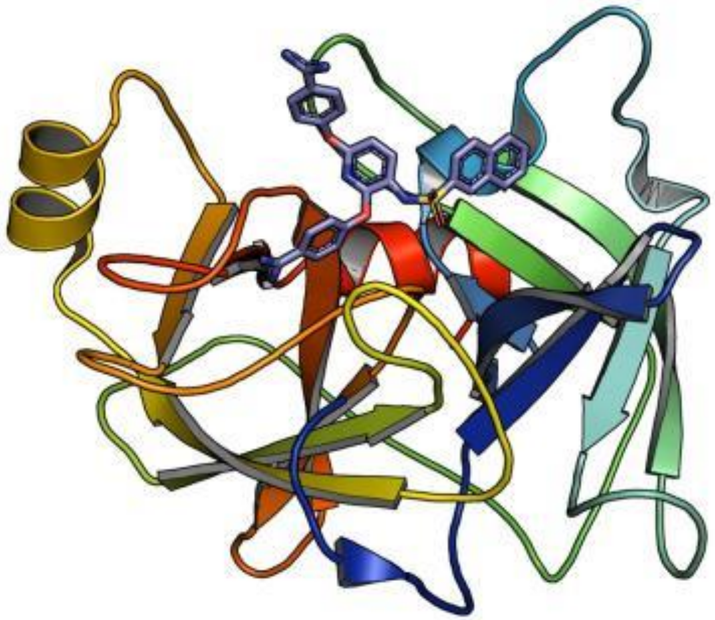
Case study 4

Structure-based design of matriptase inhibitors

Statistics										
Resolution (High)	Resolution (Low)	Cut-off Sigma (F)	Number Reflections (Observed)	Number Reflections (R-Free)	Percent Reflections (Observed)	R-Factor (Observed)	R-Work	R-Free	R-Free Selection Details	Mean Isotropic B
1.3	17.75	2	58805	2978	96.2	0.184	0.184	0.193	SHELLS	13.7





PDB ID 4JYT

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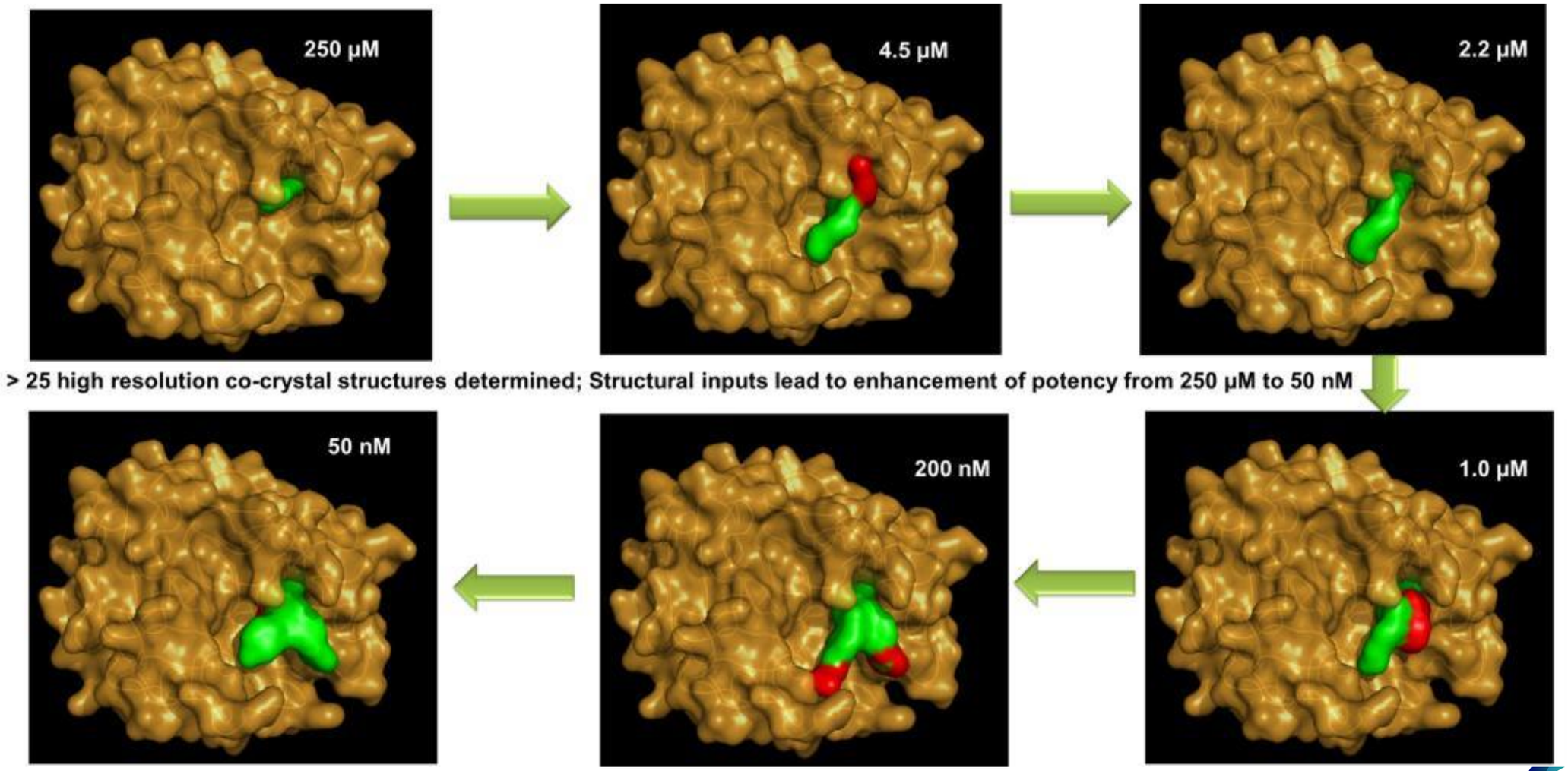
Case study 4

Structure-based design of matriptase inhibitors

FRAGMENT BASED DRUG DESIGN



> 25 high resolution co-crystal structures determined; Structural inputs lead to enhancement of potency from 250 µM to 50 nM

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Gene to Structure

Typical project format (E. coli expression example)

Phase 1 (4-6 weeks)	
Molecular biology	Construct design, codon optimization, gene synthesis (outsourced), cloning into suitable vector and sequence confirmation
Expression trials	Expression screening (8-12 conditions), one step purification and titer estimation
Phase 2 (3-6 weeks)	
Protein scale up	5-10L batch, 2-3 step purification, detailed QC including mass spec and functional test (if available)
Crystallization screening	Up to 500 conditions, using literature precedence Ligand complex generation and crystallization, soaking Cryo screening (synchrotron)
Phase 3 (3-5 weeks)	
Diffraction data collection	Complete data sets (at least two per protein/complex)
Structure solution and refinement	Molecular replacement

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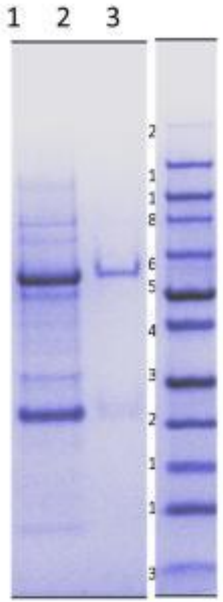
Gene to Protein: Case study

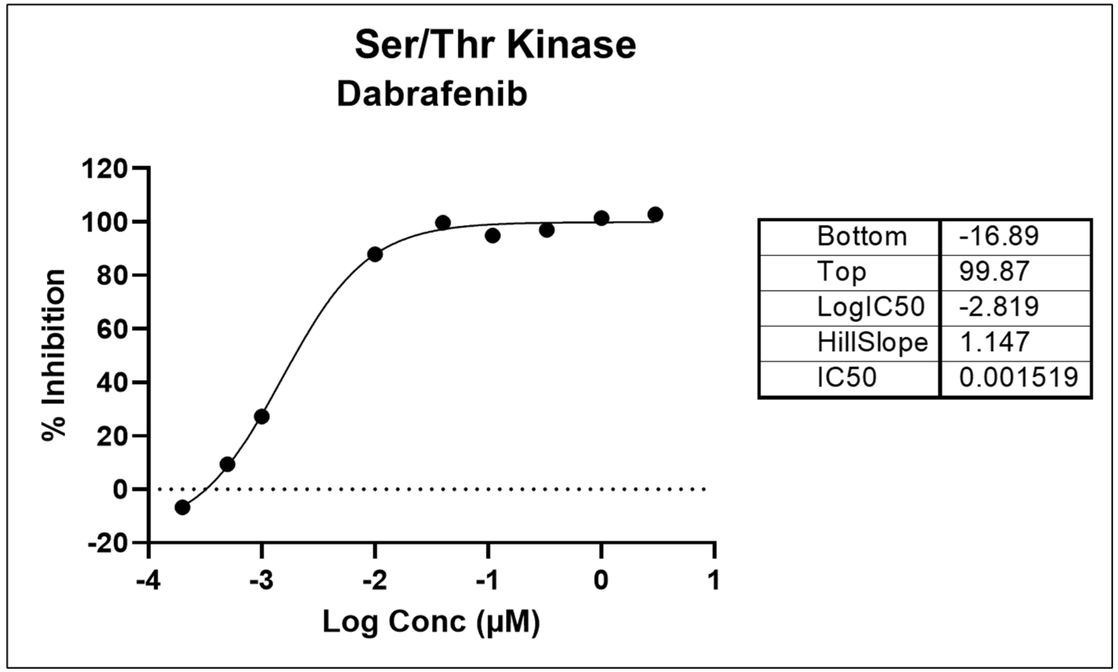
Insect cell expressed Ser-Thr kinase (Novel target)

- Difficult to express kinase target.
- Expression scouting in different expression systems with GST tag for ease of folding.
- Insect cell expression yielded good quality protein.
- Major challenge of purifying active and stable protein was overcome by working on the different conditions and methods of purification.
- Buffer optimization for stability of the target kinase



Ser/Thr kinase Truncated– 61 KDa

Gel lanes:
1. Ser/Thr Kinase Truncated- After first step
2. Ser/Thr Kinase Truncated- After final step
3. Protein Marker

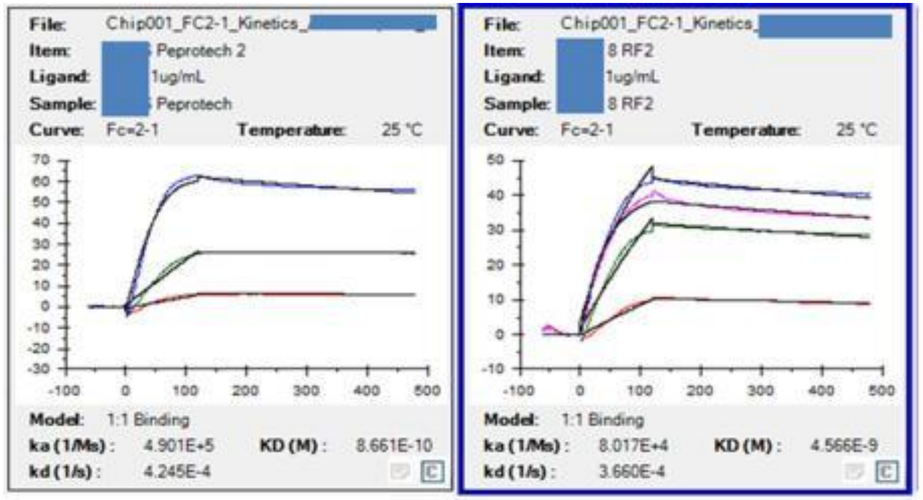


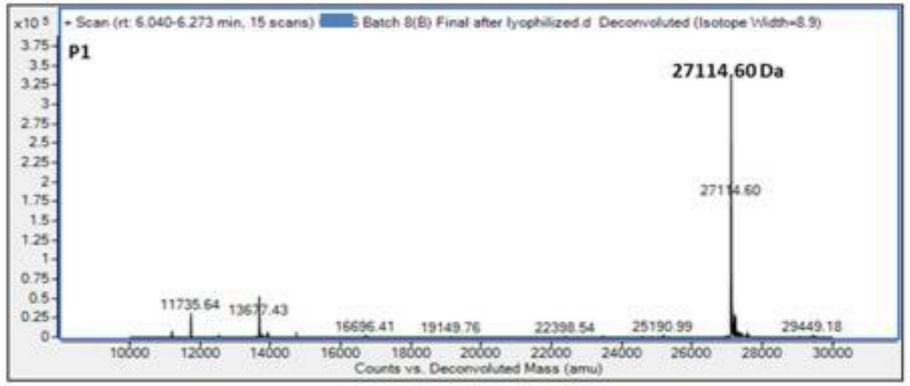
Bottom	-16.89
Top	99.87
LogIC50	-2.819
HillSlope	1.147
IC50	0.001519

Activity and inhibition reported for the first time for this Ser/Thr kinase in purified protein biochemical assay

Gene to Protein: Case study

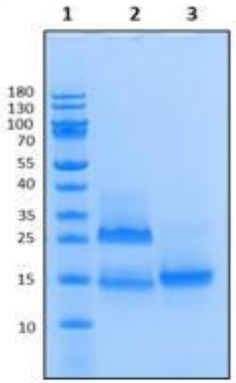
Proof of concept : process development in bacterial system





Deconvoluted Spectrum

Theoretical Mass (Da)	Observed Mass (Da)	Comment
27,390 (Dimer)	27,114	Mass confirmed with Methionine truncation



Sample details:

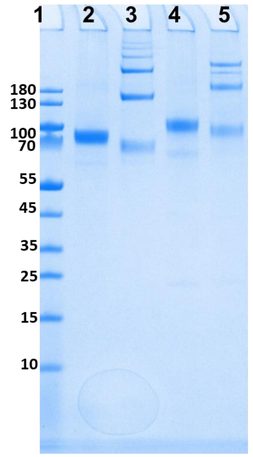
1. Pre-stained protein marker
2. Non reduced target protein
3. Reduced target protein

- *Target protein produced in bacterial system*
- *Gene cloning, expression, purification and refolding was performed in-house.*
- *All purification and refolding process was developed in-house*
- *No known precedence for purification, refolding process development and large-scale production of clinical grade material*
- *All commercial sources were for research use only*
- *A Novel SPR based method developed for QC analysis and release assay based on receptor binding of the target molecule*

Gene to Protein: Case study

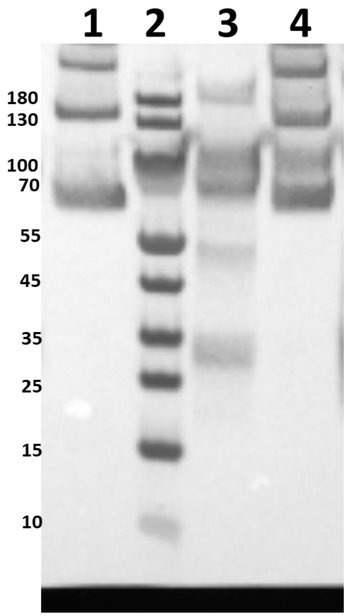
Single pass membrane protein vaccine candidate

- *Untagged target protein was expressed in insect cells along with single pass membrane associated domain*
- *Cells harvested and membrane fraction purified by detergent based washing steps*
- *Purification in presence of detergents in multi step IEX and group affinity columns*
- *Purified fractions checked with protein specific antibodies in western blot analysis*
- *Final protein formulated in desired buffer*
- *SDS-PAGE and western blot for purified sample shown.*



Sample details:

1. *Protein Marker*
2. *Pool 1- Reduced*
3. *Pool 1- Non-Reduced*
4. *Pool 2- Reduced*
5. *Pool 2- Non-Reduced*



Sample details:

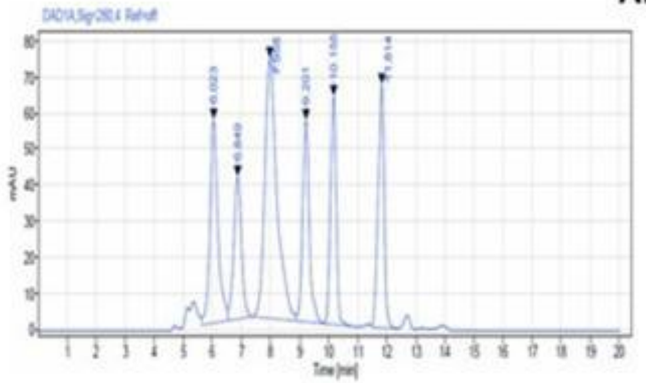
1. *Positive control*
2. *Protein Marker*
3. *Pool 1- Reduced*
4. *Pool 1- Non-Reduced*

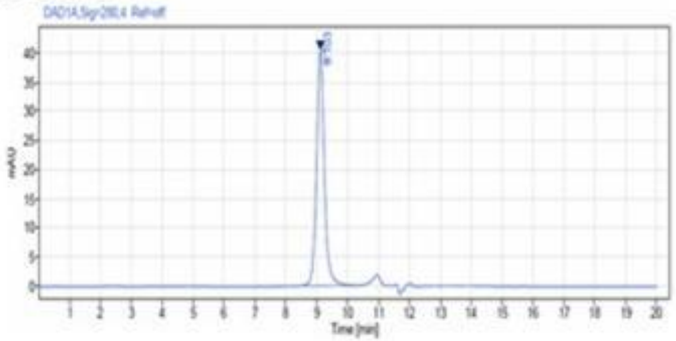
Gene to Protein: Case study

Human enzyme - oncology target

- *Target enzyme cloned expressed and purified by two step purification.*
- *Analytical QC steps done are SDS-PAGE in reducing and non-reducing conditions*
- *Analytical Size exclusion (A-SEC) run along with standards*
- *LC-MS intact mass analysis and confirmation with theoretical molecular weight calculated from the sequence*

Analytical SEC



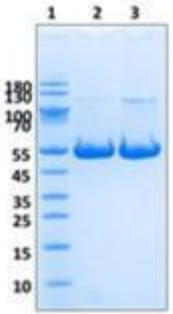


#	Name	Molecular weight (kDa)	RT (min)
1	Thyroglobulin	670	6.023
2	IgA	300	6.849
3	IgG	150	7.958
4	Ovalbumin	44	9.201
5	Myoglobin	17	10.155
6	Uridine	0.244	11.814

Signal: DAD1A,Sig=280,4 Ref=off

RT [min]	Width [min]	Area	Height	Area%	Start time [min]	End time [min]
9.103	2.01	713.80	40.37	100.00	8.255	10.265
	Sum	713.80				

SDS-PAGE



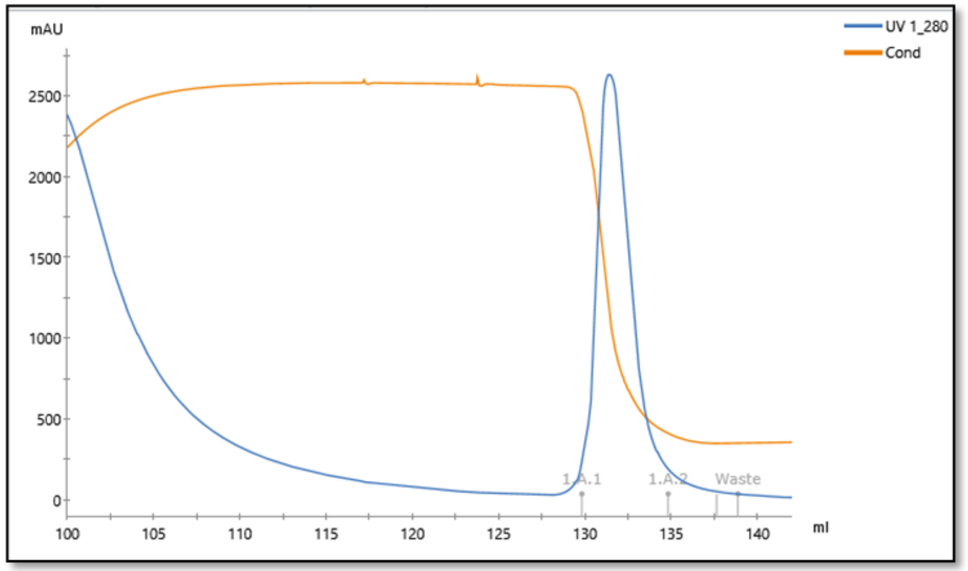
Sample details:
1. Pre-stained protein marker
2. Final sample-Reduced
3. Final sample- Non reduced

Mass Spec analysis

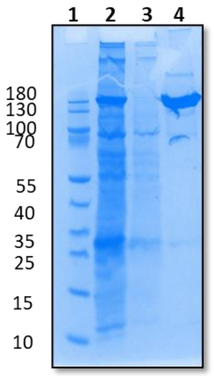
Theoretical molecular weight (Da)	Observed molecular weight (Da)	Comment
47566.16	47567	Protein confirmed

Gene to Protein: Case study

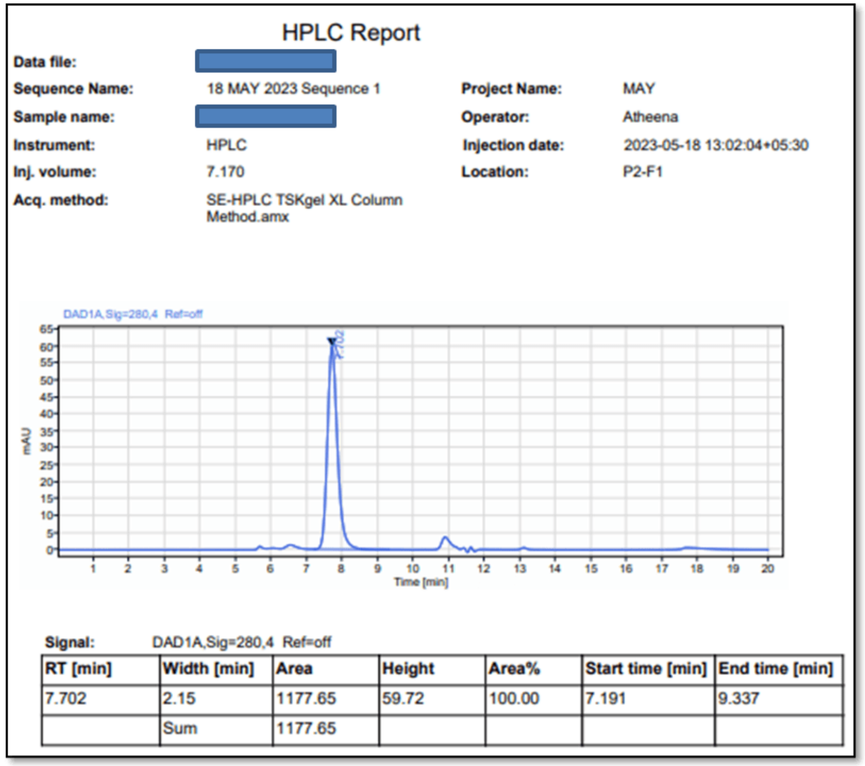
Expi CHO expressed monoclonal Antibody purification



SDS-PAGE (Non-reduced)



Lane	Samples
1	Marker
2	Load
3	Flow through
4	1A1



HPLC Report

Data file: [redacted]
Sequence Name: 18 MAY 2023 Sequence 1
Sample name: [redacted]
Instrument: HPLC
Inj. volume: 7.170
Acq. method: SE-HPLC TSKgel XL Column Method.amx
Project Name: MAY
Operator: Atheena
Injection date: 2023-05-18 13:02:04+05:30
Location: P2-F1

Signal: DAD1A,Sig=280,4 Ref=off

RT [min]	Width [min]	Area	Height	Area%	Start time [min]	End time [min]
7.702	2.15	1177.65	59.72	100.00	7.191	9.337
	Sum	1177.65				

- *Typical antibody purification involves, expression in ExpiCHO system*
- *One step or two step purification with Protein A affinity and size exclusion chromatography*
- *QC involving SE-HPLC, LC-MS, reducing and non-reducing SDS-PAGE, Endotoxin test by Endosafe® cartridges*

Gene to Protein

Typical analytical package

Sl. No.	Parameters	Read out	Typical Target profile
1	Protein concentration	Absorbance at 280 nm	
2	Purity	SDS-PAGE (reduced and non-reduced)	>95% densitometric
3	Purity	SE-HPLC	>95%
4	Identity	Intact and reduced-deglycosylayed mass (LC-MS)	20-50 ppm accuracy
5	Endotoxin level	Endosafe cartridge	<1 E.U/mg (typical)
6	Glycoform	LC-MS	
7	N-terminal sequence	Peptide mass fingerprint	
8	Western blot	Simple JESS	

Additional analytical characterization can be carried out as required (eg CD, DLS, iCE, RP-HPLC etc)

Gene to Protein

Integral membrane protein purification strategy

NanoTemper®
Prometheus NT48

- *Typical integral membrane protein extraction scheme will involve over expression, cell disruption and membrane extraction*
- *Solubilization of membrane fraction in buffers containing detergents like n-Dodecyl β-D-maltoside (DDM), n-Decyl-β-D-Maltopyranoside (DM), Octyl-beta-glucoside (OG) and Lauryldimethylamine-N-oxide (LDAO) etc., (The best detergent for protein stability can be determined after purification using nanoDSF)*
- *Ultra centrifugation of the solubilized membranes for extraction and clarification*
- *One step or two step purification by affinity, size exclusion chromatography or IEX*
- *Detergent screening for stability of membrane proteins using nano DSF*
- *Purified membrane proteins for activity assay or for structural studies*